

Build the Recommender

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can implement a deterministic recommender.

I can handle ties and no eligible options.

I can explain recommendations with their limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 33: Build the Recommender

A transparent baseline makes later improvements understandable.



THINKER MISSION

Implement a scoring table and produce ranked outputs.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

An option violates a hard duration limit but has the best score. Should it win?

Readiness for Building the data pipeline

What are the steps between raw records and a recommendation?

My readiness explanation

This week's end goal

By the end, I can implement deterministic filtering, scoring and ties for a complete catalogue case.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use build the recommender to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Build a fictional activity recommender using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name. This makes the outcome reproducible rather than dependent on input order. Assume unique names and validated numeric fields.

Apply the connected idea: Building the data pipeline

Build the validation stage of the planned chooser first. A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5. The batch stage must separately check name uniqueness. Only accepted records may reach the Week 24 selection function.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Implement a scoring table and produce ranked outputs.



PROCESS

A transparent baseline makes later improvements understandable.





OUT

OUTPUT

Implement a scoring table and produce ranked outputs.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def recommend(items, limit):  
    best = None  
    best_score = None  
    for item in items:  
        if item["minutes"] <= limit:  
            score = 2 * item["fun"] + item["learning"]  
            if (best is None or score > best_score  
                or (score == best_score and item["name"] < best)):  
                best = item["name"]  
                best_score = score
```

```

    return best

items = [{"name": "B", "fun": 3, "learning": 2, "minutes": 10},
         {"name": "A", "fun": 2, "learning": 4, "minutes": 9}]

print(recommend(items, 10))

```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: B

Evidence: Eligible score 8

Meaning: Temporary best B

Step or case: A

Evidence: Eligible score 8

Meaning: Alphabetical tie gives A

Read each row from left to right and connect the evidence to the stated rule.

The function prints nothing; the caller prints A. `best` starts as `None` to mean no eligible option found. Python short-circuits or: when `best` is `None`, later comparisons in that condition need not run. After selecting an item, `best_score` is numeric. An empty list or no eligible item returns `None`.

Trace the best-so-far record

Write a trace table with item, eligible, score and best-so-far. In the worked list, B qualifies and scores 8, so B becomes the current best. A also qualifies and scores 8. The scores tie, so the written alphabetical rule changes the best to A. Starting with A gives the same final answer. If no item qualifies, `best` stays `None`; do not invent a winner or quietly relax the limit.

Apply the connected idea: Building the data pipeline

```

def valid_item(row):
    if type(row) is not dict:
        return False

    if type(row.get("name")) is not str or not row["name"]:
        return False

    for key, maximum in [("minutes", 60), ("fun", 5), ("learning", 5)]:

```

```

        value = row.get(key)

        if type(value) is not int or not 0 <= value <= maximum:

            return False

    return True

print(valid_item({"name": "A", "minutes": 20, "fun": 4, "learning": 2}))
print(valid_item({"name": "B", "minutes": 10, "fun": "4", "learning": 2}))

```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Raw rows

Evidence: Check fields and types

Meaning: Accepted or rejected

Step or case: Accepted rows

Evidence: Check unique names

Meaning: No ambiguous identifiers

Step or case: Unique valid rows

Evidence: Apply Week 24 selector

Meaning: Suggestion plus counts

Read each row from left to right and connect the evidence to the stated rule.

The worked calls print True then False. Short-circuit checks keep absent or wrongly typed fields away from later operations. This Boolean helper does not explain the reason; add reason codes in the reporting stage or use a validation result record. Do not describe a range failure as a no-match outcome.

Trace the hand-off between stages

Think of validation and selection as two desks. At the first desk, each record is checked for required fields, exact types and allowed ranges; a batch check also checks unique names. Only accepted records go to the second desk, where the time limit and scoring policy are applied. A record can be valid yet too long to qualify. Report rejected and ineligible separately so a reader knows which decision removed an item.

Connect the picture and the explanation

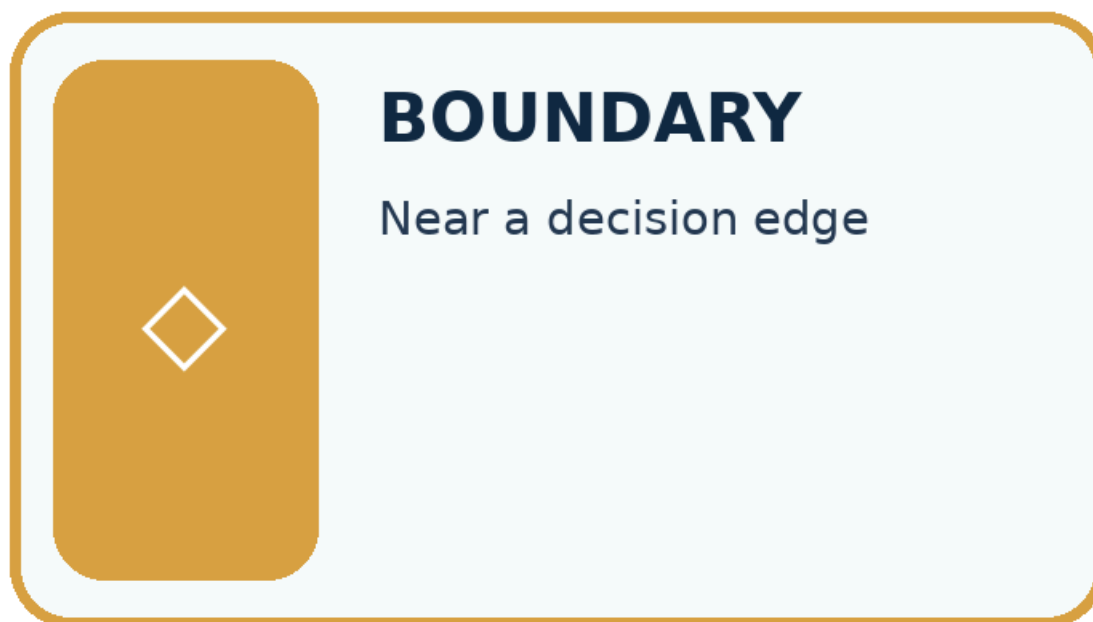
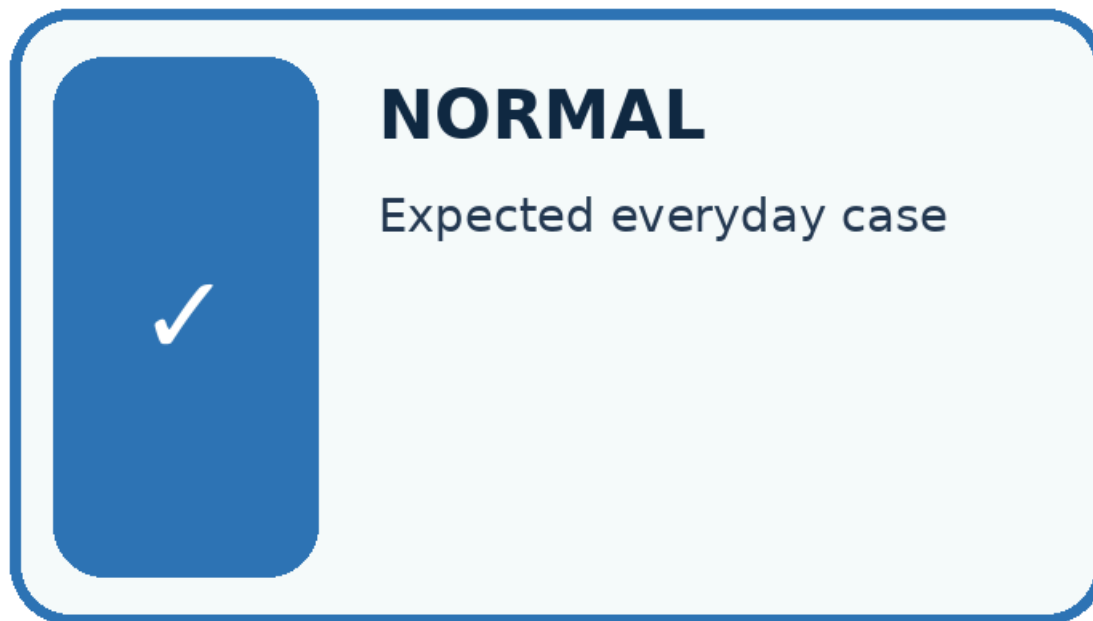
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

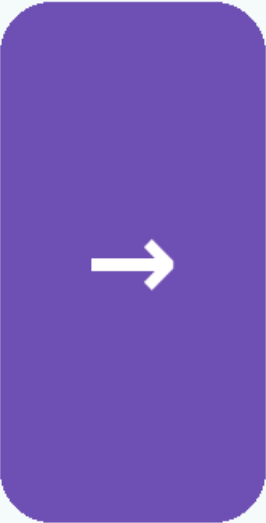


NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Change D's pages to 91, leaving every other field unchanged.

Prediction: D becomes ineligible, so A wins with score 2.

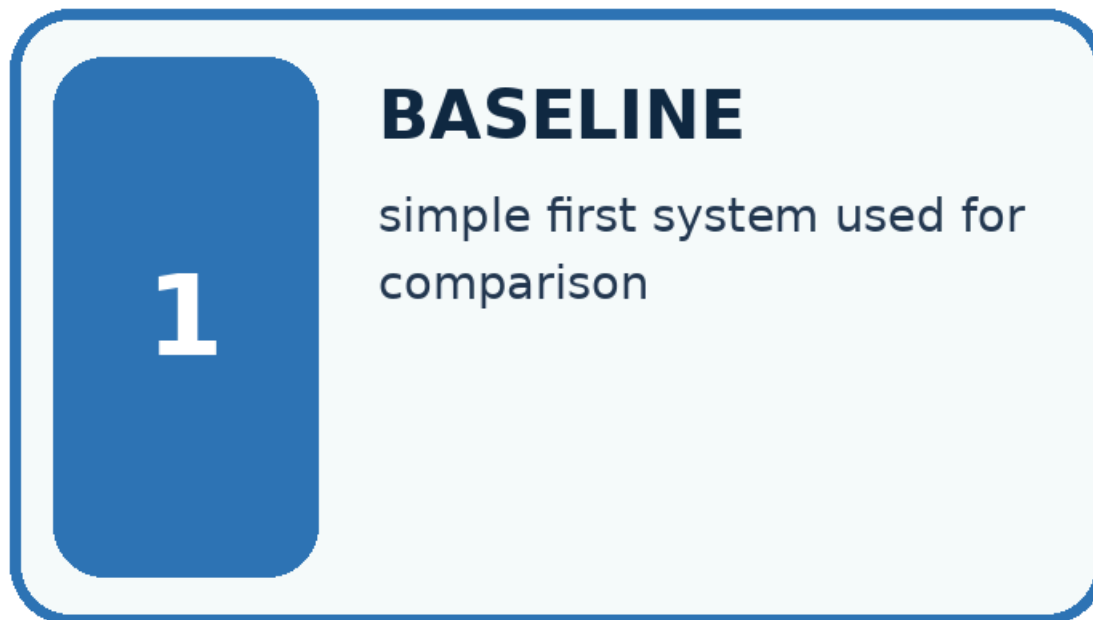
Reason: The strict excess over the 90-page maximum changes filtering before ranking.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

CANDIDATE

item being scored

3

WEIGHTED TOTAL

sum of feature matches
times weights

4

EXPLANATION

reason tied to visible evidence

Word · My own example · Not this / boundary

Word: baseline

My own example: A simple reference method used to judge whether a more complex method adds value.

Not this / boundary: _____

Word: candidate

My own example: An item or proposed answer being considered before a final selection.

Not this / boundary: _____

Word: weighted total

My own example: A sum formed after multiplying each contribution by its assigned weight.

Not this / boundary: _____

Word: explanation

My own example: An account connecting a result to the mechanism, calculation or evidence that supports it.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Recommender

Meaning: A procedure that suggests options using a stated policy

Word: Tie

Meaning: Two options have the same comparison score

Word: Deterministic

Meaning: The same inputs and rules produce the same result

Terms for Building the data pipeline

Term · Meaning

Term: Stage

Meaning: One part of a processing pipeline

Term: Validation function

Meaning: A function that decides whether an input meets a schema

Term: Separation of concerns

Meaning: Keeping different responsibilities in distinct parts

Term: Rejection reason

Meaning: An explanation of why a record was not accepted

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 33 • WORKED EVIDENCE

- 1 Weights: genre=3, difficulty=2, length=1.
- 2 Compute every candidate's terms and total.
- 3 Filter hard constraints, sort remaining totals high to low.
- 4 Output top choice, score breakdown, one limitation, and a tie rule.

1. Weights: genre=3, difficulty=2, length=1.
2. Compute every candidate's terms and total.
3. Filter hard constraints, sort remaining totals high to low.
4. Output top choice, score breakdown, one limitation, and a tie rule.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Predict the result if the list order is reversed, and if the limit is 8.

Apply the connected idea: Building the data pipeline

Predict validity for minutes 0, fun 0, learning 5 with name A, then for minutes 61.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Build the Recommender

A fictional book chooser filters a request before comparing preference scores.

The facts we will use

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

Filtering before scoring

	Pages	Preference score	Status
A	90	2	eligible
B	70	1	eligible
C	60	3	subject fails
D	80	3	eligible

Read a cell using both its row and column labels.

Build the Recommender

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Filter

A,B,D meet subject and page requirements; C fails subject.



2. Compute scores

A=2; B=1; D=3.



3. Choose best

Return D with both preference matches recorded.



4. Deterministic output

Reordering the input rows must not change D's unique winning score.

CHANGE ONE THING

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Filter

What happens and why: A,B,D meet subject and page requirements; C fails subject.

Step: 2. Compute scores

What happens and why: A=2; B=1; D=3.

Step: 3. Choose best

What happens and why: Return D with both preference matches recorded.

Step: 4. Deterministic output

What happens and why: Reordering the input rows must not change D's unique winning score.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded. The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an explanation, not just the final ID.

What this example does not establish

Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A recommendation is transparent if it displays only the final score.
Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Write a one-sentence explanation of X winning without calling it objectively best.

Apply the connected idea: Building the data pipeline

Each stage should have a clear input and output contract. A program can be correct at one stage but wrong at their connection, such as passing rejected rows into selection. Keep acceptance and rejection counts in the report.

Two valid rows have the same name but different ratings. Should individual validation alone permit selection?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Updating best on score $\geq$ best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties. Test both orders and a no-match case after the repair.

Apply the connected idea: Building the data pipeline

Calling the selector on raw data can crash or produce a result from invalid ratings. Route data through validation first. Duplicate names require a batch check even when each row individually passes; block the ambiguous batch for review instead of arbitrarily keeping one.

Practise with less help: Handle no match

Use workbook W2 for the connected example “Implementing a recommender”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Building the data pipeline

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Explain why B does not win after D is excluded even though it has fewer pages.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Build the editable score table, hand-check one row, run two profiles, and compare system ranking with independent recalculation.

Build the editable score table, hand-check one row, run two profiles, and compare system ranking with independent recalculation.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

This is the complete baseline: inputs, scoring, filter, ranking, explanation, and abstain rule.

THIS WEEK'S CHECKPOINT This is the complete baseline: inputs, scoring, filter, ranking, explanation, and abstain rule.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Save the recommender code, contract and this full trace; test the same records in different input orders.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Implement a scoring table and produce ranked outputs.

CHALLENGE Implement a scoring table and produce ranked outputs.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

The question

Compute the winner and evidence to display.

Worked explanation

A wins at 3; B=2; C=1. Show flags, formula, scores and explanation; declare tie policy.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Build and expose the scoring calculation

Evidence or stage · Value or result

Evidence or stage: A, genre and length

Value or result: 3

Evidence or stage: B, genre only

Value or result: 2

Evidence or stage: C, length only

Value or result: 1

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Compute the winner and evidence to display.

A wins at 3; B=2; C=1. Show flags, formula, scores and explanation; declare tie policy.

14. Try a new case independently

Independent case: Changed independent case

Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated. Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

This is the complete baseline: inputs, scoring, filter, ranking, explanation, and abstain rule.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering. Show why other books are excluded.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.